Paper Structure

Abstract

A concise summary of the project, highlighting the motivation, the standardized dataset format for causal reasoning, the dual-agent LLM pipeline (answering and evaluation), and the main findings from experiments.

1. Introduction

Introduce the motivation and problem statement:

- Why causal reasoning matters in AI and machine learning.
- The challenge of evaluating LLMs on causal tasks in a consistent and interpretable way.
- The main **objective** of the project: to create a standardized framework that allows any causal dataset to be transformed, processed, and evaluated with minimal user effort.

2. Related Work

Provide an overview of relevant research and technologies:

- Previous work on **causal reasoning in machine learning** and **causal inference benchmarks**.
- Studies or benchmarks that inspired your project (e.g., existing causal datasets like Cause-Effect pairs, COPA, or others).
- LLM-related work on **explanation generation** and **evaluation**.
- References to tools, frameworks, or methods you relied on (datasets, evaluation metrics, prompt engineering techniques, etc.).
  This situates your contribution in the broader field.

3. Solution Architecture

Describe the design and approach in detail:

- **Overall architecture**: the two-agent pipeline (Answering LLM + Evaluation LLM).
- **Dataset standardization**: explain the mapping process that converts arbitrary causal datasets into the common schema (context, question type, choices, label, explanation).
- **Models used**: describe which LLMs were applied and how they were prompted or configured.
- Provide diagrams or structured explanations of how data flows through the system.

4. Experiments

Explain the experimental setup:

- Which datasets were used and how they were transformed.
- Details of the experimental procedure (how many examples, how many runs, etc.).
- Prompting strategies, model parameters, and any baselines considered.
- Enough detail so another researcher could replicate the experiments.

5. Results

Present and discuss findings:

- Quantitative metrics (accuracy, agreement rates, evaluation confidence, etc.).
- Comparative results: across datasets, across models, or compared to existing published results.
- Interpret results with tables and figures, showing trends and insights.
- Discuss key observations (e.g., where explanations were strong vs. weak).

6. Limitations

Acknowledge the boundaries of the study:

- Dependence on the quality of ground-truth explanations.
- Subjectivity in explanation evaluation.
- Constraints due to model costs, dataset diversity, or scalability.

7. Qualitative Examples

Showcase concrete examples:

- Good cases: where the model provided correct answers with strong causal reasoning.
- Failure cases: where the model's explanation was incorrect, shallow, or misleading.
- These examples illustrate strengths and weaknesses of the methodology in practice.

8. Conclusion

Summarize the contributions:

- Standardization of causal datasets into a unified schema.
- A two-agent LLM pipeline for answering and evaluating causal reasoning tasks.
- Key results and insights.
  Reinforce how the project advances causal evaluation of LLMs.

9. Future Work

Outline next steps:

- Extending to more diverse datasets.
- Improving the evaluation agent with better semantic or causal alignment metrics.
- Integrating automatic scoring methods (e.g., embedding similarity, causal graph alignment).
- Scaling the framework for larger studies and benchmarking.